MILITARY SPECIFICATION

PANELS, GROUND DETECTOR, GENERAL

SPECIFICATION FOR

1. SCOPE

1.1 This specification covers the general requirements for alternating current (a.c.) and direct current (d.c.) ground detector panels.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-B-613 - Brass, Leaded and Non-Leaded, Flat Products (Plate, Bar, Sheet and Strip).

MILITARY

MIL-M-14 - Molding Plastics, and Molded Plastics, Thermosetting.
MIL-I-631 - Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-S-901 - Shock Tests, H.I. (High Impact) Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917 - Electric Power Equipment Basic Requirements for Naval Shipboard Use.
MIL-E-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard.
MIL-T-7928 - Terminals, Lug: Splices, Conductor, Crimp Style, Copper, General Specification For.
MIL-P-15024 - Plates, Identification - Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-P-15037 - Plastic Sheet, Laminated Thermosetting, Glass-Cloth Melamine-Resin.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and packing of.
MIL-P-24395/1 - Ground Detector Panel, 120-Volt (Symbol 1005.1) and 450-Volt (Symbol 1005.2) A.C. Three Phase for 60 or 400 Hertz, Dripproof.
MIL-P-24395/2 - Ground Detector Panel, 120-Volt (Symbol 1005.3) and 240-Volt (Symbol 1005.4) D.C. Dripproof.
MIL-I-45208 - Inspection System Requirements.
MIL-T-55164 - Terminal Boards, Molded, Barrier Screw and Stud Type.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-278 - Welding and Inspection of Machinery, Piping and Pressure Vessels for Ships of the U.S. Navy.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28-Screw Thread Standard for Federal Services.

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(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL FIRE PROTECTION ASSOCIATION
ASA CI - National Electric Code.

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Mass., 02110.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to beginning production, a sample panel of each symbol number under a contract or order shall be inspected as specified in 4.2 (see 6.3).

3.1.1 The first article sample panel, indicator lights, fuseholders, switches, terminal boards, resistors and wiring shall be furnished complete as specified on the individual specification sheets.

3.2 Specification sheets. The individual panel requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern.

3.3 Materials. Materials shall be as specified (see 6.2) and shall conform to MIL-E-917. Material not definitely specified shall be of the quality best suited for the purpose intended.

3.3.1 Restrictive material. Flammable or explosive material, or material which can produce toxic or suffocating fumes shall not be used.

3.3.2 Insulation material. Laminated plastics shall be type GME in accordance with MIL-P-15037. Compounds used for molding mounting bases, parts, and so forth, shall conform to Type MAI-60 or Type MMI-30 of MIL-M-14.

3.4 Milling, countersinking or tapping of metal parts shall be accomplished before plating or finish is applied.

3.5 Threaded parts. Unless otherwise specified, threads shall be right hand, Class 2 conforming with unified and American National Screw Threads as specified in Handbook H-28.

3.5.1 Self-tapping devices and sheet metal screws shall not be used in the assembly of the panels.

3.6 Edges and corners. All external projecting edges and sharp corners normally exposed, shall be rounded.

3.7 Welding, brazing and similar processes. Welding, brazing, and similar processes used in manufacture of Ground Detector Panels shall be in accordance with MIL-STD-278.

3.8 Terminal boards. Terminal boards shall conform to MIL-T-55164 and as specified on the individual specification sheet.

3.9 Lugs and terminals. Lugs and terminals shall be in accordance with MIL-T-7928.

3.10 Wire and terminal marking. Internal wiring of equipment shall be marked by stamping on wire terminals or where terminals are too small to permit stamping, wire markers of synthetic resin tubing, Type F, Grade A of MIL-I-631 or fiber tags are acceptable. Terminal blocks shall be equipped with marking strips. The marking strips shall be permanently marked at each terminal to agree with the corresponding number on the wiring diagram.

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3.11 Wiring space. Adequate cabling space shall be provided for both power input cables as well as internal cabling. The cabling spaces shall conform to the requirements of National Electric Code Handbook.

3.11.1 Interior wiring. All interior wiring shall be furnished by the manufacturer, using stranded copper wire of the type specified by the individual specification sheet. Connections at screw terminals shall be made with crimp-type connectors in accordance with MIL-T-7928. Connectors shall be reliably prevented from turning unless such turning does not decrease the creepage and clearance distances (see 3.20.1). Interior wiring shall be connected as shown in the wiring diagram on the individual specification sheet. All wiring shall be neatly formed into groups which shall be laced, tied or clamped in a manner that provides support and prevents chafing of the wire insulation due to vibration and shock. There shall be no splices in the wire and all connections shall be made at the terminals of the devices or at terminal blocks.

3.12 Stress relief.

3.12.1 Metals. All metals used in fabrications and assembly shall be treated or heat treated to prevent deterioration or failure due to stresses or other conditions resulting from working, forming, and similar processes during the fabrication of the panels.

3.12.2 Plastics. Adequate measures shall be taken in molding or processing plastics to insure that stress buildup does not occur or is treated to relieve these stresses to prevent deterioration or failure of the part or assembly.

3.12.3 "Stress relieved" shall be stamped on outside bottom of the enclosure in 1/4-inch high letters with ink of a grade similar or equal to "clear print opaque" manufactured by Phillips Process Co., Inc., Rochester, New York.

3.13 Vibration. The panels shall be designed to withstand Type I vibration test of MIL-STD-167, without mechanical damage or malfunction during or after the vibration test.

3.14 Shock. The panels shall withstand the shock tests for Grade A, class 1 equipment as specified in MIL-S-901 without loosening of parts or damage or distortion of sufficient magnitude to impair continued operation of the panel.

3.15 Corrosion. The panels shall withstand the salt spray test specified in 4.5.7.

3.16 Enclosures. The enclosure of the ground detector panel shall conform to the requirements specified in the applicable specification sheet. If a material is not specified in the applicable enclosure specification sheet, brass conforming to QQ-B-613 shall be used.

3.16.1 The enclosures shall be thoroughly cleaned of all soldering flux and other corrosive agents. Cleaning agent shall be neutralized after cleaning. Cleaning shall be in accordance with MIL-E-917 (not primed or painted).

3.16.2 Holes for stuffing tubes or box connectors. When specified (see 6.2), holes for stuffing tubes or box connectors shall be cut or drilled. The holes shall be smooth and true without any distortion of the box.

3.17 Wiring diagram. Wiring diagram in a holder shall be installed inside the enclosure in accordance with the requirements of MIL-E-2036.

3.18 Degree of enclosure. The degree of enclosure for ground detector panels shall be as specified on the applicable specification sheet.

3.19 Size limitations. Panels shall be as small and light as practical, and shall be sufficiently rugged to withstand the shock test (see 3.14). Mounting dimensions shown on individual specification sheets shall be maintained and maximum overall dimensions shall not be exceeded.

3.20 Electrical requirements.

3.20.1 Clearances and creepage. Minimum electrical creepage and clearances shall be as specified in the National Electric Code Handbook. These clearances are applicable to machined laminated plastic bases or parts. Clearances applicable to molded bases or supports may conform to the requirements of MIL-E-917 provided mold finishes are not disturbed.

3.20.2 Dielectric withstanding voltage. The ground detector panels shall be capable of withstanding voltage test of 4.5.2.

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3.20.3 Grounding. Detector panels shall be designed to be free of shock hazard to the maximum practical extent. The design and construction of the panels shall be such that all exposed metal parts or other electrically conductive material (panel cover and box, switches, indicator lights, fuseholders) are at ground (ships hull) potential at all times. The exposed metal portions both interior and exterior of electrical parts such as switches, indicator lights and fuseholders shall be in physical contact with or electrically connected to the grounded panel. A ground wire shall be provided between the cover and the box as shown on the individual specification sheet so that the cover remains grounded when removed from the box.

3.20.4 Insulation resistance. The ground detector panels shall be capable of withstanding the insulation resistance test of 4.5.5. The insulation resistance shall be not less than 200 Megohms.

3.20.5 Heat. When subjected to the heat test of 4.5.6, the temperature rise in current carrying parts (bus connection points, and terminals) shall not exceed 45°C. above an ambient room temperature of 25°C. and the air temperature within the panel shall not exceed 65°C.

3.21 Drawings. Drawings shall consist of manufacturers drawings, (certified by the cognizant Government inspector as complying with the requirements of this specification). The certification shall be documented on the drawing as a matter of record. Drawings shall carry manufacturers numbers and the number of this specification. (Formal approval and validating signatures for manufacturers drawings will not be required nor will NAVSHIPS drawing numbers be assigned.)

3.21.1 Manufacturer's drawings shall be furnished and shall include the following:

(a) Outline, not necessarily to scale, of the front; side, cross sectional, and top views of the panels, giving dimensions and other essential details. In the interest of clarity, views in addition to those specified may be shown.
(b) List of material.
(c) Descriptive data of the equipment, which shall include the degree of enclosure.
(d) Weight of the equipment, center of gravity, and mounting dimensions.
(e) A list of notes and references pertinent to the basic design of the equipment.
(f) A list of notes pertinent to the tests conducted in accordance with section 4 of this specification.
(g) Title block.

3.21.2 Drawing distribution. The manufacturer, upon being advised that the first article sample has successfully passed the tests of this specification shall immediately place on record a set of prints of the drawing used for manufacturing the parts with the following:

(a) One set for file with the local Government inspector.
(b) One set for file with the Naval Ship Engineering Center, Department of Navy, Hyattsville, Maryland 20782.
(c) One set for file with the Contracting Officer, Ships Parts Control Center, Mechanicsburg, Pennsylvania 17055.
(d) One set for file with the Commander, Defense General Supply Center, Richmond, Virginia 23212.

3.22 Identification plates. The ground detector panels shall have an identification plate in accordance with type B of MIL-P-15024. The markings shall be engraved on the cover in accordance with the individual specification sheets and shall be depressed in the metal to provide permanent identification. The identification plate shall contain the following information:

(a) Name of the item.
(b) Voltage.
(c) Symbol number.
(d) Military part number.
(e) Specification sheet number.
(f) Federal stock number.
(g) Contract number.
(h) Manufacturer's (Prime contractor) name or identification.

3.22.1 Identification plates shall be designed for severe service in accordance with MIL-P-15024/5.

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3.23 Painting. Painting and protection against corrosion shall be in accordance with MIL-E-917, or as specified (see 6.2).

3.24 Workmanship. The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The inspection system which the supplier is required to maintain, as provided in the inspection clause of the contract or order, shall be in accordance with MIL-I-45208 (see 6.5).

4.2 First article inspection. The sample(s) for First article inspection shall be subjected to the examination and test specified in Table I.

Table I - First article inspection.

Inspection	Requirement paragraph	Test paragraph
Examination	3.2, 3.19, 3.23, 3.24	4.4
Dielectric	3.20.2	4.5.2
Insulation	3.20.4	4.5.5
Heat	3.20.5	4.5.6
Enclosure	3.18	4.5.1
Shock	3.14	4.5.3
Vibration	3.13	4.5.4
Salt spray	3.15	4.5.7
Creepage and clearance	3.20.1	4.5.8

4.2.1 Redesign of panel. Redesign of a panel, once the manufacturer has established the panel design as a matter of record, shall require resubmission of a sample for first article inspection.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of the examination of 4.4, the effectiveness of enclosure test of 4.5.1 and the dielectric test of 4.5.2.

4.3.1 Sampling.

4.3.1.1 Lot. For the purpose of sampling, a lot shall be considered to be the identical number of panels of each symbol number on contract or order.

4.3.1.2 Sampling for examination and effectiveness of enclosure and dielectric tests. A random sample of panels of each symbol shall be selected in accordance with Table II from each inspection lot and shall be subjected to the examination and tests specified in 4.3. Any panel containing one or more defects and any panel failing in one or more test shall not be offered for delivery. If the number of such defective panels in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall not be offered for acceptance.

Table II - Sampling for quality conformance inspection.

Number of panels in inspection lot	Number of panels in sample	Acceptance number	Rejection number
2 to 10	all	--	--
11 to 15	10	0	1
16 to 25	13	0	1
26 to 40	17	0	1

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Table II - Sampling for quality conformance inspection (Cont'd).

Number of panels in inspection lot	Number of panels in sample	Acceptance number	Rejection number
41 to 65	22	0	1
66 to 110	28	1	2
111 to 180	35	1	2
181 to 300	45	2	3
301 to 500	55	2	3
501 and over	70	3	4

4.4 Examination. The ground detector panels shall be examined to determine conformance with the requirements of this specification and the applicable specification sheet which do not require testing and that materials, design, construction, dimensions, marking and workmanship have been met. Alinement of parts shall be observed to determine that the parts are in good working order.

4.5 Tests.

4.5.1 Effectiveness of enclosure. Panels shall be tested to determine the effectiveness of the enclosure in accordance with MIL-STD-108.

4.5.2 Dielectric. Panels shall be subjected for 1 minute to a dielectric test voltage, the effective potential of which is twice rated voltage, plus 500 volts, applied between points of opposite polarity and between live parts and ground.

4.5.3 Shock. Panels shall be subjected to the high impact shock tests specified for grade A, class 1 equipment of MIL-S-901 to determine conformance with 3.14. There shall be no readjustment during testing.

4.5.4 Vibration. Panels shall be subjected to the vibration tests up to and including the frequency range of 15 Hertz for type I equipment as specified in MIL-STD-167 to determine conformance with 3.13. The panels shall perform its principal functions during and after the vibration tests.

4.5.5 Insulation resistance. Insulation resistance of electrical circuits following the dielectric tests shall be measured with a 500 volt D.C. megometer between electrical conductors and between electrical conductors and ground at approximately room temperature 75°F. (25°C.) and at relative humidity of approximately 50 percent.

4.5.6 Heat test. The sample(s) shall be subjected to heat testing. The heat testing shall be made under simulated conditions equivalent to normal operating conditions employing maximum rated voltage, maximum rated load and rated frequency. The test shall be conducted continuously until the temperatures have remained constant for 1 hour.

4.5.7 Salt spray. The sample(s) shall be subjected to the salt spray tests specified in method 101, condition A of MIL-STD-202. The salt solution concentration shall be 20 percent.

4.5.8 Creepage and clearance. Creepage and clearance distances between current carrying parts and the panel shall be measured to determine conformance with 3.20.1.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.

5.1.1 Panels.

5.1.1.1 Preservation and packaging. Preservation and packaging which may be the supplier's commercial practice shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

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5.1.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier, at lowest rate, and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations applicable to the mode of transportation and may conform to the supplier's commercial practice.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include item name, Military Part number, symbol number, Federal stock number, manufacturer's part number, contract or order number, manufacturer's name and destination.

5.1.2 Onboard repair parts. Onboard repair parts shall be preserved and packaged by level A, packed by level C and marked by level A and C respectively in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment. The requirements, and levels of preservation, packaging, packing and marking for shipment shall be as specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation and packaging, packing and marking. The equipment and accessories, and onboard repair parts shall be preserved and packaged by level A or C; packed by level A or B and marked in accordance with MIL-E-17555.)

6. NOTES

6.1 Intended use. The ground detector panels are for use in surface ships for the detection of ground faults in electrical systems, such as lighting systems and 400 Hertz systems that have distribution panels as main distribution centers. Distribution panels would not normally have space for mounting the ground detector lights as would a switchboard, therefore the surface mounted ground detector panel would be mounted adjacent to the distribution panel.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Number of individual specification sheet, symbol number and Military Part number.
(c) Painting, if other than in accordance with MIL-E-917 (see 3.23).
(d) If holes are required for stuffing tubes (see 3.16.2).
(e) Level of preservation, packaging, packing and marking required, if other than as specified in 5.1 (see 5.2).

6.2.1 Bid data. The manufacturer, or his representative, shall for each symbol of each panel on an order, furnish a suitable sketch showing internal bussing arrangement (circuit arrangement), overall size and weight of panel and mounting dimensions.

6.3 First article inspection.

6.3.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Management control system documents. The following management control system document should be included on DD form 1660:

(a) MIL-I-45208 (see 4.1.1).

Preparing activity:
Navy - SH
(Project 6110-N075)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO (If "YES", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD, FORM 1426 APR 63 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 0101-807-2000

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